Notebook 3 — Chapter 3.3

Selection Bias, Adverse Selection & Information Economics

20 worked examples — conditioning on execution, the winner's curse, trading against asymmetric information — calibrated for Stage 2 Finals market-making intuition

A. Conditioning on Trade Execution

Example 1 — The Core Identity, Made Concrete

True value V is equally likely to be any of $\$10, \$20, \ldots, \$100$ (10 values). Your resting ask is \$70; a buy order executes against it iff $V \geq 70$. Find $\mathbb{E}[V \mid \text{trade executed}]$ and compare to $\mathbb{E}[V]$.

Solution. Values ≥ 70: $\{70, 80, 90, 100\}$, mean $= \dfrac{70+80+90+100}{4} =$ $\boxed{\$85}$. $\mathbb{E}[V] = \dfrac{10+\cdots+100}{10} = \$55$.

⋆ *Intuition:* $\mathbb{E}[V \mid \text{executed}] \neq \mathbb{E}[V]$ isn't a subtle correction — here it's a \$30 gap. A fill is never a random draw from the population; it's a draw from the *sub*-population that satisfies your execution condition.

Example 2 — The Mirror Image: Resting Bid

Same population. Your resting bid is \$40; a sell order hits it iff $V \leq 40$. Find $\mathbb{E}[V \mid \text{executed}]$.

Solution. Values ≤ 40: $\{10, 20, 30, 40\}$, mean $=$ $\boxed{\$25}$.

▶ *Intuition:* you bought at \$40 but the average true value of fills is only \$25 — an average \$15 overpayment per fill, purely from the selection effect of *who chooses to sell to you*. This is adverse selection's entire mechanism in one number.

Example 3 — The Continuous Version

$V \sim \text{Unif}(0, 100)$. Ask $=$ \$70. Find $\mathbb{E}[V \mid V \geq 70]$.

Solution. For $\text{Unif}(0, 100)$ truncated above a:

$$\mathbb{E}[V \mid V \geq a] = \frac{a+100}{2} = \frac{70+100}{2} = \boxed{\$85}$$

⋆ *Speed trick:* a truncated uniform is still uniform (on $[a, 100]$), so its mean is just the midpoint of the truncated range — confirms Example 1's discrete answer exactly, continuous or discrete makes no difference here.

Example 4 — Quantifying the Adverse-Selection Loss

$V \sim \text{Unif}(0, 100)$. Resting bid $= b$. Find $\mathbb{E}[V \mid V \leq b]$ and the average overpayment on a fill.

Solution.

$$\mathbb{E}[V \mid V \leq b] = \frac{b}{2}$$

$$\text{Overpayment} = b - \frac{b}{2} = \boxed{\frac{b}{2}}$$

▶ *Intuition:* the loss is *exactly half your bid*, for any uniform V on $[0, 100]$ — a strikingly simple closed form. The higher you bid, the larger your expected overpayment on every single fill, not just in rare bad cases.

Example 5 — Capping the Loss

Using Example 4's formula, find the bid b such that the average overpayment on a fill is capped at \$5.

Solution.

$$\frac{b}{2} = 5 \;\Rightarrow\; b = \boxed{\$10}$$

⋆ *Intuition:* to keep adverse-selection loss under \$5 on a $\text{Unif}(0, 100)$ contract, you must bid at only \$10 — a full \$40 below the population's naive midpoint of \$50. This is the quantitative seed of "defensive spreads" in Section C: one-sided selection risk alone forces quotes far from fair value, before inventory or informed flow even enter the picture.

B. The Winner's Curse

Example 6 — Baseline: Updating on Your Own Signal

A jar's true value V is \$80 or \$120, equally likely. Each bidder privately observes a signal S: correct w.p. 0.8, equal to the *other* value w.p. 0.2. Bidder 1 sees $S_1 = 120$. Find $\mathbb{E}[V \mid S_1 = 120]$ — *no* competition yet.

Solution. Bayes:

$$\mathbb{P}(V = 120 \mid S_1 = 120) = \frac{0.8(0.5)}{0.8(0.5) + 0.2(0.5)} = \frac{0.4}{0.5} = 0.8$$

$$\mathbb{E}[V \mid S_1 = 120] = 120(0.8) + 80(0.2) = \boxed{\$112}$$

⋆ *Intuition:* this is ordinary signal-updating, nothing auction-specific yet — your private information alone pulls your estimate from the \$100 prior toward your signal. Section B's whole point starts in Example 7: winning the auction adds *more* information on top of this.

Example 7 — The Winner's Curse, Fully Derived

Same setup, 2 bidders, signals conditionally independent given V. Bidder 1 wins (bids $=$ signal, ties coin-flipped) iff $S_1 \geq S_2$. Given $S_1 = 120$, find $\mathbb{E}[V \mid S_1 = 120, \text{win}]$.

Solution. Joint probabilities: $\mathbb{P}(S_1 = 120, S_2 = 80) = \frac{4}{25}$, $\mathbb{P}(S_1 = 120, S_2 = 120) = \frac{17}{50}$, summing to $\mathbb{P}(S_1 = 120) = \frac{1}{2}$.

$$\mathbb{P}(S_2 = 80 \mid S_1 = 120) = \frac{8}{25}, \quad \mathbb{P}(S_2 = 120 \mid S_1 = 120) = \frac{17}{25}$$

Within each branch: $\mathbb{E}[V \mid S_1{=}120, S_2{=}80] = 100$ (signals disagree $\Rightarrow$ no net info beyond the prior!); $\mathbb{E}[V \mid S_1{=}120, S_2{=}120] = \frac{2000}{17} \approx 117.65$.

$$\mathbb{P}(\text{win} \mid S_1{=}120) = \frac{8}{25}(1) + \frac{17}{25}\left(\frac{1}{2}\right) = \frac{33}{50}$$

$$\mathbb{E}[V \mid S_1{=}120, \text{win}] = \frac{\frac{8}{25}(100) + \frac{17}{25} \cdot \frac{1}{2} \cdot \frac{2000}{17}}{33/50} = \frac{1200}{11} \approx \boxed{\$109.09}$$

▶ *Full-circle intuition:* three numbers, same signal, three different answers: naive bid $=$ \$120 (just your signal), signal-adjusted $=$ \$112 (Example 6), win-adjusted $=$ \$109.09 (here). **Winning is itself bad news** — you're more likely to win when your opponent's signal was low, which is evidence V might genuinely be lower. This gap is the winner's curse, computed exactly, not asserted.

Example 8 — The Correct Bid

Using Example 7, what should a rational bidder actually bid, instead of naively bidding their raw signal of \$120?

Solution. Bid $\mathbb{E}[V \mid S_1 = 120, \text{win}] \approx$ $\boxed{\$109.09}$, *not* \$120 and not even \$112.

⋆ *Intuition:* rational bid-shading means conditioning your bid on the hypothetical *event of winning itself* — a fixed-point-flavored idea (your bid depends on what winning would tell you, which depends on your bid). In practice, bidders shade down from their raw signal by an amount that grows with the number of competitors (Example 9) and the informativeness of signals.

Example 9 — More Bidders, Deeper Curse

Repeat Example 7's computation with $n = 3$ bidders (still $S_1 = 120$, same signal channel). Find $\mathbb{E}[V \mid S_1 = 120, \text{win}]$ and compare to the $n = 2$ result.

Solution. Exact computation (summing over both other bidders' signals):

$$\mathbb{E}[V \mid S_1{=}120, \text{win}, n{=}3] = \frac{3952}{37} \approx \boxed{\$106.81}$$

versus \$109.09 at $n = 2$ (Example 7) and \$112 with no competition (Example 6).

▶ *Intuition:* more competitors means winning is a *stronger* statement — beating 2 rivals instead of 1 is rarer and more informative that their signals (and so, weakly, V) were on the low side. This is exactly why common-value auctions with many bidders see far more aggressive bid-shading than two-bidder ones.

Example 10 — The Curse Is a Fill, in Disguise

Explain why Example 2's bid-fill loss (\$15 overpayment) and Example 7's auction winner's curse (\$120→\$109.09) are the *same* phenomenon.

Solution. In both cases, an event ("my bid gets hit" / "I win the auction") is *positively correlated with the counterparty having better information that the true value is unfavorable to me*. Formally, both are instances of

$$\mathbb{E}[V \mid \text{Trade Executed}] \neq \mathbb{E}[V]$$

where "Trade Executed" $=$ "my resting order was the best price available," exactly the condition that defines winning a common-value auction.

⋆ *Intuition:* a resting limit order IS a standing bid in a continuous common-value auction against the entire market's collective information

— every fill you get is a won auction, and every won auction carries the winner's curse.

Example 11 — How Signal Quality Shapes the Curse

Recompute Example 7's gap $\big(\mathbb{E}[V \mid S_1{=}120] - \mathbb{E}[V \mid S_1{=}120, \text{win}]\big)$ for a less informative channel, accuracy $a = 0.6$, versus the original $a = 0.8$.
Solution. General baseline: $\mathbb{E}[V \mid S_1{=}120] = 80 + 40a$. At $a = 0.6$: baseline $= 104$, win-adjusted $= \$102.70$, gap $\boxed{\approx \$1.30}$. At $a = 0.8$: baseline $= 112$, win-adjusted $= \$109.09$, gap $\boxed{\approx \$2.91}$.
▶ *Intuition:* more informative private signals (a closer to 1) create a *larger* winner's curse gap, not a smaller one — sharper private information makes disagreement among bidders more meaningful, so winning (implicitly, "my rivals disagreed with me downward") reveals more. (Bonus: as $a \to 1$ the gap eventually shrinks back toward 0, since perfectly informative signals leave no residual uncertainty to exploit — the curse is worst at intermediate information quality.)

C. Trading Against Asymmetric Information

Example 12 — The Glosten–Milgrom Ask

$V \in \{80, 120\}$ equally likely. A fraction $\alpha = 0.4$ of order flow is *informed* (knows V exactly; buys iff $V{=}120$, sells iff $V{=}80$); the rest is noise (buys/sells 50/50 regardless of V). Find the break-even ask, $\mathbb{E}[V \mid \text{buy order}]$.
Solution. $\mathbb{P}(\text{buy} \mid V{=}120) = \alpha + (1-\alpha)(0.5) = 0.7$; $\mathbb{P}(\text{buy} \mid V{=}80) = (1-\alpha)(0.5) = 0.3$.

$$\mathbb{P}(V{=}120 \mid \text{buy}) = \frac{0.7(0.5)}{0.7(0.5) + 0.3(0.5)} = 0.7$$

$$\text{ask} = 120(0.7) + 80(0.3) = \boxed{\$108}$$

⋆ *Intuition:* a market maker who ignores informed flow entirely and quotes the naive \$100 fair value loses money on average to informed buyers — the ask must rise above \$100 purely to break even against the *possibility* that any given buy order is informed.

Example 13 — The Symmetric Bid, and the Pure Information Spread

Same setup ($\alpha = 0.4$). Find the break-even bid, $\mathbb{E}[V \mid \text{sell order}]$, and the resulting spread.
Solution. By symmetry:

$$\text{bid} = 80(0.7) + 120(0.3) = \boxed{\$92}$$

$$\text{spread} = 108 - 92 = \boxed{\$16}$$

▶ *Intuition:* with no informed flow at all ($\alpha = 0$), both formulas collapse to ask $=$ bid $=$ \$100 — zero spread. The *entire* \$16 spread here exists purely to compensate for adverse selection; this is a clean separation from Ch. 3.2's inventory-driven spread, which exists even with zero informed flow.

Example 14 — The Closed-Form Spread, and Why It Widens

Derive the general formula for the Glosten–Milgrom ask as a function of α (symmetric 50/50 prior, values L, H), and evaluate at $\alpha = 0.8$.
Solution. Following Example 12's algebra with general α: $\mathbb{P}(H \mid \text{buy}) = \frac{1+\alpha}{2}$, so

$$\text{ask} = \frac{H+L}{2} + \alpha \cdot \frac{H-L}{2} = \text{FV} + \alpha \cdot \frac{H-L}{2}$$

At $\alpha = 0.8$ ($H{=}120, L{=}80$): $\boxed{\text{ask} = 100 + 0.8(20) = \$116}$, bid $= \$84$, spread $= \boxed{\$32}$ (double Example 13's, since α doubled).
⋆ *Speed trick:* the spread is *exactly linear* in α: spread $= \alpha(H - L)$. Memorize this closed form — it turns every "find the defensive spread" question in this chapter into one substitution.

Example 15 — Solving for α from an Observed Spread

You observe a competitor quoting a \$24 spread on the same \$80/\$120 contract. Using Example 14's formula, back out their assumed α.
Solution. spread $= \alpha(H - L) = \alpha(40) = 24$

$$\alpha = \boxed{0.6}$$

▶ *Intuition:* exactly like Ch. 3.2's quotes revealing a competitor's σ, a Glosten–Milgrom-style spread reveals how toxic a maker believes their flow is — reading competitors' spreads is a real way to estimate market-wide informed-trading intensity.

Example 16 — When the Prior Isn't 50/50

Same mechanics, $\alpha = 0.5$, but now $\mathbb{P}(V{=}120) = 0.3$ (asymmetric prior). Find the ask, bid, and spread — the shortcut formula no longer applies directly.
Solution. $\mathbb{P}(\text{buy} \mid 120) = 0.75$, $\mathbb{P}(\text{buy} \mid 80) = 0.25$. $\mathbb{P}(\text{buy}) = 0.3(0.75) + 0.7(0.25) = 0.4$.

$$\mathbb{P}(120 \mid \text{buy}) = \frac{0.225}{0.4} = 0.5625 \Rightarrow \text{ask} = 120(0.5625) + 80(0.4375) = \boxed{\$102.50}$$

Symmetrically: $\mathbb{P}(\text{sell}) = 0.6$, $\mathbb{P}(80 \mid \text{sell}) = 0.875 \Rightarrow$ bid $= \boxed{\$85}$. Spread $= \boxed{\$17.50}$.
⋆ *Intuition:* Example 14's clean linear formula was a special case of a symmetric 50/50 prior — with an asymmetric prior you must redo full Bayes on each side separately; the ask and bid no longer sit symmetrically around the \$92 naive fair value ($0.3 \times 120 + 0.7 \times 80$).

D. Capstones

Example 17 — The General Glosten–Milgrom Formula

Derive ask $= \mathbb{E}[V \mid \text{buy}]$ symbolically for general prior $p = \mathbb{P}(V{=}H)$, informed fraction α, and values L, H.
Solution. $\mathbb{P}(\text{buy} \mid H) = \frac{1+\alpha}{2}$, $\mathbb{P}(\text{buy} \mid L) = \frac{1-\alpha}{2}$.

$$\mathbb{P}(H \mid \text{buy}) = \frac{p(1+\alpha)}{p(1+\alpha) + (1-p)(1-\alpha)}$$

$$\text{ask} = H \cdot \mathbb{P}(H \mid \text{buy}) + L \cdot \big(1 - \mathbb{P}(H \mid \text{buy})\big)$$

▶ *Intuition:* every numeric example in Section C (12, 13, 14, 16) is one substitution into this single formula — $p = 0.5$ collapses it to Example 14's clean linear form; any other p requires the full fraction, as in Example 16.

Example 18 — Sequential Updating: Information Can Cancel

Using Example 12's setup ($\alpha = 0.4$), a buy order arrives first (posterior $\mathbb{P}(V{=}120) = 0.7$, per Example 12). A sell order arrives next, using this 0.7 as the new prior. Find the posterior after both trades.
Solution. $\mathbb{P}(\text{sell} \mid 120) = 0.3$, $\mathbb{P}(\text{sell} \mid 80) = 0.7$. With prior 0.7/0.3:

$$\mathbb{P}(\text{sell}) = 0.7(0.3) + 0.3(0.7) = 0.42$$

$$\mathbb{P}(V{=}120 \mid \text{buy, then sell}) = \frac{0.7(0.3)}{0.42} = \boxed{0.5}$$

⋆ *Intuition:* the posterior reverts *exactly* to the original 50/50 prior — a buy and a subsequent sell, with equal informed-fraction on each side, perfectly cancel in information content. Price discovery from order flow is a running, reversible Bayesian update, not a one-way ratchet.

Example 19 — Stacking Information Risk on Inventory Risk

$V \in \{80, 120\}$ (so FV $= 100$, $\sigma^2 = 400$), $\alpha = 0.4$ (Example 12's \$8 info half-spread). A maker also holds $q = 5$, $\gamma = 0.01$. Find the final two-sided quote combining *both* effects (Ch. 3.2's reservation price + this chapter's info half-spread).
Solution. Reservation price:

$$r = \text{FV} - q\gamma\sigma^2 = 100 - (5)(0.01)(400) = \boxed{\$80}$$

Info half-spread from Example 12/13: ±\$8 around fair value. Apply it around r instead:

$$\text{bid} = 80 - 8 = \boxed{\$72}, \qquad \text{ask} = 80 + 8 = \boxed{\$88}$$

▶ *Intuition:* inventory risk (Ch. 3.2) and adverse-selection risk (this chapter) are *additive, independent* skews — inventory moves the *center*, informed-flow risk sets the *width* around that center. Compute each separately, then combine; never conflate the two sources.

Example 20 — Full Capstone: Every Risk Source, One Quote

$V \in \{60, 140\}$ equally likely (FV $= 100$, $\sigma^2 = 1600$), $\alpha = 0.3$. Maker is short $q = -4$, $\gamma = 0.005$. Find the final bid/ask.
Solution. Info half-spread (Example 14's formula): $\alpha \cdot \frac{H-L}{2} = 0.3(40) =$

12, so info ask/bid would be 112/88 around FV. Reservation price:

$$r = 100 - (-4)(0.005)(1600) = 100 + 32 = \boxed{\$132}$$

Final quote, info half-spread around r:

$$\text{bid} = 132 - 12 = \boxed{\$120}, \qquad \text{ask} = 132 + 12 = \boxed{\$144}$$

⋆ *Full-circle intuition:* the entire chapter sequence in one quote — a high-variance contract ($\sigma^2 = 1600$) makes the short position's inventory skew enormous ($32, dwarfing the $12 info-driven half-spread), while the informed-flow risk sets how wide to quote around that shifted center. Large position sizes in volatile, information-sensitive contracts can push quotes dramatically far from naive fair value — exactly the stacked-risk intuition a Stage 2 market-making round is built to test.

One-tool-tree reminder: every result in this chapter is $\mathbb{E}[V \mid \text{selection event}] \neq \mathbb{E}[V]$, applied three ways — to a simple execution threshold (Section A), to winning a common-value auction (Section B, the same conditioning applied to "my signal was highest"), and to Bayesian order-flow inference against a known fraction of informed traders (Section C). Section D is just these three, plus Ch. 3.2's reservation price, all stacked on one quote.